Week 5 Assignment Louis Desgrée: Implementation and Testing of the SmoothWall Firewall in a Virtualized Environment

Introduction

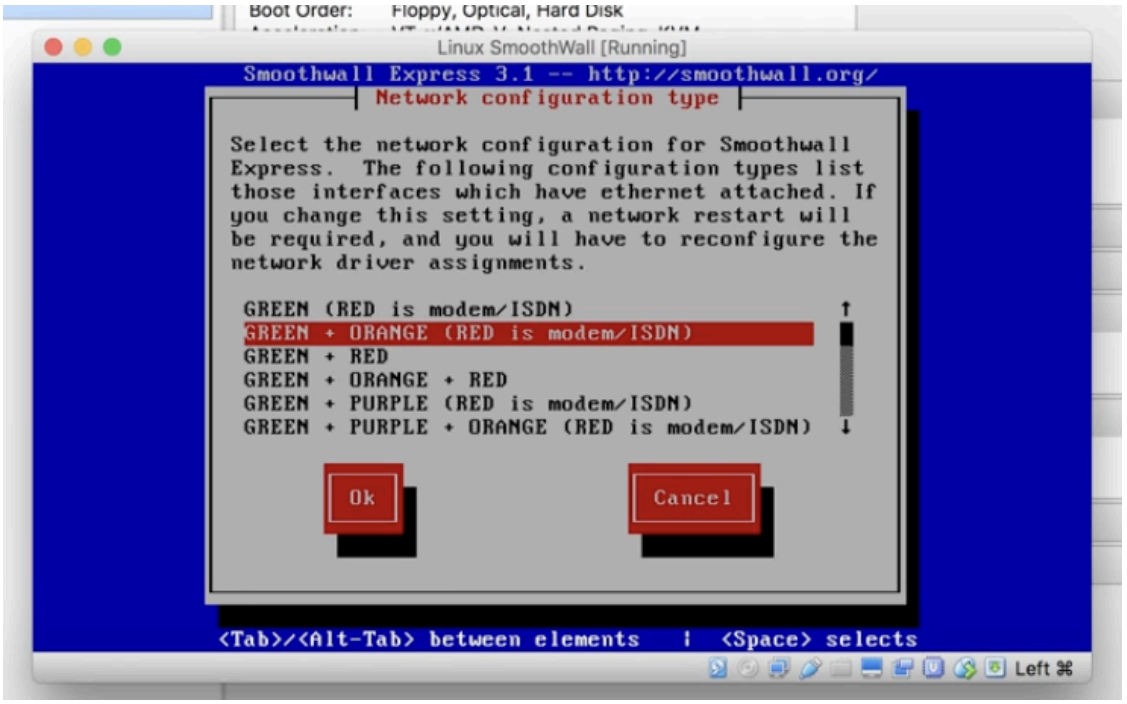
This document presents a detailed account of the steps taken to implement and test the SmoothWall firewall within my virtual network environment. The objective of this project was to integrate a robust firewall solution into an architecture segmented by security zones, ensuring that each zone was appropriately isolated and monitored. Throughout the process, I captured several screenshots to document key stages, which will be referenced at appropriate points in this report.

Downloading and Setting Up the Virtual Machine

I began by downloading the SmoothWall 3.1 installation package from the official SourceForge repository. Recognizing the importance of proper hardware configuration in a virtualized setting, I created a new virtual machine using my preferred virtualization platform. A critical decision during the setup was to configure the virtual hard drive as IDE instead of SCSI, as recommended by the installation guidelines. This precaution was essential because previous attempts with a SCSI configuration had led to issues during subsequent restarts, potentially causing the firewall to become unresponsive.

Additionally, I configured three network adapters corresponding to the designated security zones:

- **Red**: Untrusted network (e.g., external internet connection)
- **Orange**: Partially trusted devices (e.g., DMZ)
- **Green**: Internal network (trusted machines)

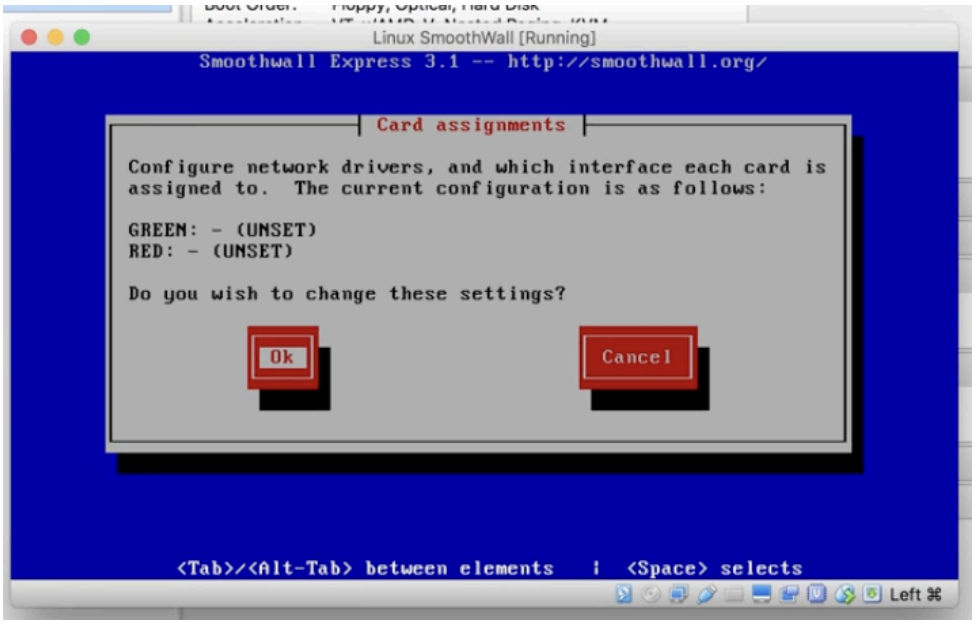


SmoothWall Installation Process

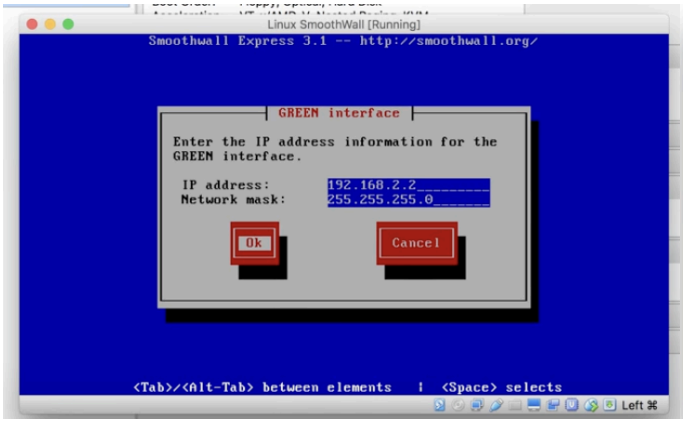
Following the virtual machine setup, I proceeded with the installation of SmoothWall. The installation wizard provided clear instructions that guided me through each step, from partitioning the virtual disk to configuring basic network settings.

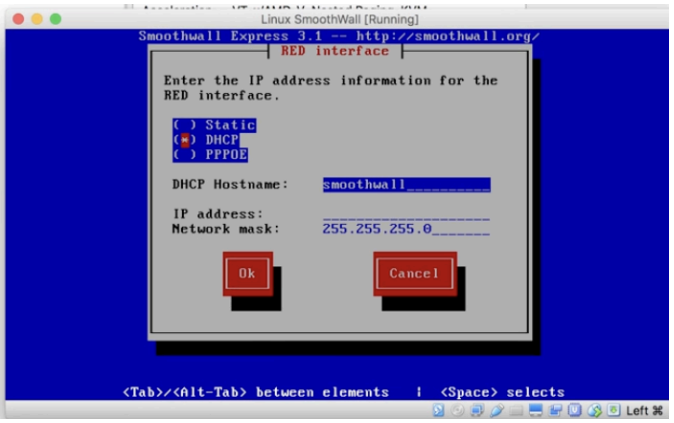
During the network configuration phase, I assigned roles to each interface:

- **Green interface**: Assigned an internal IP address (192.168.2.2/24)
- **Red interface**: Configured to use DHCP for dynamic IP assignment



Once the installation was successfully completed, I rebooted the system and connected a dedicated client machine to the Green (internal) network to access the firewall's web interface.

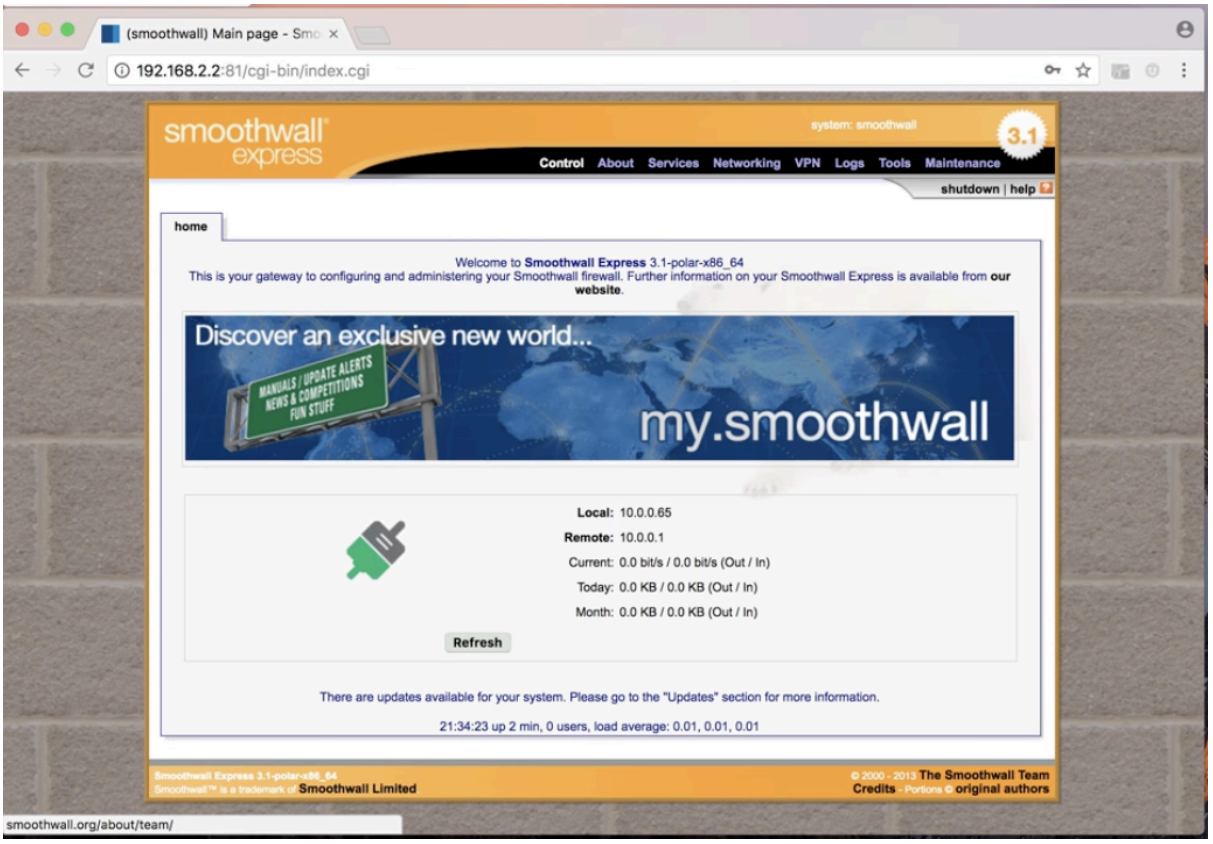




Accessing the SmoothWall Web Interface

After configuring the network, I attempted to access the SmoothWall web interface from a browser running on a Kali Linux virtual machine. However, I encountered a security protocol issue related to TLS when accessing the firewall. To resolve this, I adjusted the browser's minimum TLS version in the about:config settings, which allowed the connection to proceed without error.

Once I successfully logged into the firewall, I captured a screenshot of the login confirmation screen.



Testing and Verification

To verify the correct functionality of the firewall, I conducted several tests:

- **Ping tests** to confirm connectivity between different zones.
- **Firewall rule verification** to ensure appropriate traffic filtering.
- **Log analysis** to check if blocked traffic attempts were properly recorded.

The firewall's logs accurately reflected the attempted connections and corresponding actions.

Conclusion

The hands-on experience of implementing and testing the SmoothWall firewall has deepened my understanding of network segmentation and security configuration. The process highlighted the importance of meticulous virtual machine configuration—especially the setting of the hard drive type—and demonstrated effective troubleshooting strategies for dealing with TLS-related issues. This project has not only reinforced best practices in firewall implementation but also equipped me with practical skills applicable in real-world network security environments.

References

https://sourceforge.net/projects/smoothwall/files/SmoothWall/3.1/

https://support.mozilla.org/en-US/kb/about-config-editor-firefox

https://www.virtualbox.org/manual/ch05.html